

1 ■ Eigenvalues and Eigen Vectors of Two by Two Matrices

Exercise 1.1. Find the eigenvalues and eigen vectors of the following matrices:

(a) $\begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$

(b) $\begin{pmatrix} -1 & 2 \\ -7 & 8 \end{pmatrix}$

(c) $\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$

(e) $\begin{pmatrix} 10 & -9 \\ 4 & -2 \end{pmatrix}$

2 ■ Eigenvalues and Eigen Vectors of Three by Three Matrices

Exercise 2.1. Find the eigenvalues and eigen vectors of the following matrices:

(a) $\begin{pmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 3 & 0 & -5 \\ 1/5 & -1 & 0 \\ 1 & 1 & -2 \end{pmatrix}$

(c) $\begin{pmatrix} -2 & 0 & 1 \\ -6 & -2 & 0 \\ 19 & 5 & -4 \end{pmatrix}$

(d) $\begin{pmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{pmatrix}$

(e) $\begin{pmatrix} 5 & 6 & 2 \\ 0 & -1 & -8 \\ 1 & 0 & -2 \end{pmatrix}$